

NAME(S):

Fitch-Margoliash Exercise

Consider four gene sequences, (A, C, D, E), having the following distance matrix:

	A	C	D	E
A	-	-	-	-
C	39	-	-	-
D	39	18	-	-
E	41	20	10	-

1. Use UPGMA to determine kinship between the four sequences. What can you use as an outgroup?
2. Using the three most related sequences, figure out appropriate branch lengths using the Fitch-Margoliash algorithm and draw the resulting *unrooted* scaled tree.
3. Use the outgroup along with the Fitch-Margoliash algorithm to create a phylogram: a *rooted* scaled tree with appropriate branch lengths for the three most related sequences.